AFC–7609

Sub. Code
1BCAA2

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Applications – Allied

OFFICE AUTOMATION

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the use of Word art?

2. Define Footer.

3. What are the features available in Excel?

4. What do you by Absolute Address in Excel?

5. What do you mean by an Animation effect?

6. What are the views available in Power Point?

7. Define Database.

8. What is the use of Form in Access?

9. What are the parts of Email? Give an example.

10. What is WWW?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the various editing operations in Word.

Or

(b) Write down the steps to insert page numbers to a document.

12. (a) Explain any five mathematical functions in Excel with example.

Or

(b) Explain the steps to create a line graph in Excel.

13. (a) Explain the steps to add a Table to a slide.

Or

(b) Explain the features of Power Point.

14. (a) Write down the steps using query to display details of the students whose Height is greater than 168 cm.

Or

(b) Explain the steps to link the tables.

15. (a) Explain the anatomy of Email.

Or

(b) Write short notes on search engines.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the steps in Mail Merge in Word.

17. Explain the steps to create Bar Graph in Excel.

18. Explain the steps to create a slide presentation with sound effect.

19. Explain the steps to create a Report.

20. Explain :

    (a) Managing Email account

    (b) Uses of Internet.

AFC–7453

Sub. Code
1BCA2C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Application

PROGRAMMING IN C++

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the basic concepts of Object Oriented Programming?

2. Define token. Give an example.

3. How will you define member functions?

4. Define Constructor.

5. What is an Abstract class?

6. What is a Pointer? Give example.

7. Define File. What are the uses of File?

8. What is the purpose of tellg( ) and seekg() functions?

9. What are the advantages of a Template?

10. Define Exception.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write a program to print the Prime numbers between 100 and 500 using while loop.

Or

(b) Explain switch statement with an example.

12. (a) Explain static data member and static member functions with an example.

Or

(b) Explain Dynamic constructor with a program.

13. (a) Explain with a program how pointers can be assigned to a derived classes.

Or

(b) Explain the various C++ stream classes.

14. (a) Explain various file opening modes in C++ with example.

Or

(b) Discuss Random access file operation.

15. (a) Explain user defined templates with an example.

Or

(b) Explain any five rules for handling Exceptions.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the operators in C++ with example.

17. Write a C++ program to add two matrices by passing object and returning object.

18. Explain Multilevel Inheritance with a program.

19. Write a C++ program to search a employee record in a file using his mobile number.

20. Write a C++ program to swap two integers, two float numbers and two double numbers using function template.

AFC–7454

Sub. Code
1BCA3C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Application

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the views of data?

2. Define constraint.

3. What do you mean by Atomic domains?

4. What do you mean by Data Dictionary?

5. What is meant by Interquery Parallelism?

6. What do you mean by Homogeneous Database?

7. Define Schema. Give an example.

8. Define Role.

9. Write any Two DCL commands.

10. Define Package.

Part B $(5 \times 5 = 25)$

Answer **all** questions by choosing either (a) or (b).

11. (a) Explain the Database System applications.

Or

(b) Explain the different types of Database Users.

12. (a) Explain the features of good Relational database designs.

Or

(b) Explain the Database design process.

13. (a) Explain the Client-Server architecture.

Or

(b) Explain the Distributed transactions.

14. (a) Explain with an example how sequences are created and deleted?

Or

(b) Explain how will you create View? Give an example.

15. (a) Explain FOR-LOOP command with an example.

Or

(b) Explain Triggers with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain

    (a) E-R Diagrams

    (b) Transaction Management.

17. Explain Decomposition using multivalued dependencies with an example.

18. Describe the Network types.

19. Explain any Ten DML commands with example.

20. Write a procedure to check whether the given string is palindrome or not a palindrome.

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AFC–7455

Sub. Code
1BCA4C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Computer Application

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write any two benefits of OOP.

2. What do you mean by Type Casting?

3. What is meant by type conversion? Give one example.

4. Write any two mathematical functions used in java.

5. How will you add variables to methods?

6. Define Interface.

7. What is meant by synchronization?

8. Define Exception.

9. What is the difference between Applet and Application?

10. Write the syntax of the method which is used to draw an Arc.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the features of java.

Or

(b) Explain the java data types with example.

12. (a) Explain the operators in Java with example.

Or

(b) Write a Java program to print the Armstrong numbers between 100 and 800.

13. (a) Write a java program to find the smallest number and its location in an array.

Or

(b) Explain how will you access interface variables.

14. (a) Explain package with a an example.

Or

(b) Explain multiple catch statements with an example.

15. (a) Explain the steps to design a web page using HTML.

Or

(b) Explain with an example how will you display numerical values in an Applet?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain :

    (a) JVM

    (b) Java Environment.

17. Write a java program to print the numbers between 100 and 500 that are divisible by 5 and not divisible by 7 and 10 using for loop.

18. Explain method overloading with a program.

19. Write a java program to create a thread by extending thread class.

20. Write a Java program to draw rectangles in different locations.

AFC–7456

Sub. Code
1BCA5C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Applications

VISUAL PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define variable. Give an example.

2. What is an array?

3. List any four properties of textbox control.

4. How to indexing in a list box control?

5. What is a clipboard?

6. Write any two graphics methods.

7. What is MDI?

8. What is parent and child menu?

9. What is the purpose of ADO data control?

10. Write any four SQL statement.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the collection objects.

Or

(b) Distinguish between subroutines and functions.

12. (a) Write about combo box control.

Or

(b) Write about scroll bar and slider control.

13. (a) Explain the properties of picture control.

Or

(b) Discuss the optimization issues.

14. (a) Explain the properties and methods of common dialogs control.

Or

(b) Explain how tree structure work.

15. (a) Discuss the ADO object model.

Or

(b) Explain about data report.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain any two looping statements with example.

17. Describe menu edition with suitable example.

18. Write note on :
    (a) Line
    (b) Shapes
    (c) Circle method
    (d) Co-ordinate system.

19. Explain the properties and methods of Rich Text Box.

20. Describe the visual data manager.

AFC–7457

Sub. Code
1BCA5C2

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Application

COMPUTER SYSTEM ARCHITECTURE AND DESIGN

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a register?

2. Define Instruction code, and operation code.

3. Define Machine language and Assembly language.

4. Draw the syntax of Memory-Reference Instruction.

5. What are the uses of Stack?

6. Write any four characteristics of RISC computer.

7. Draw the Hardware for Signed-Magnitude Addition and Subtraction.

8. Write the steps for floating point multiplication.

9. What is meant by loading of reference?

10. Which concept is used in associative memory?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about Register transfer in detail.

Or

(b) Explain about Instruction cycle.

12. (a) Explain about program loops in detail.

Or

(b) Explain about the codes and assembly, machine languages.

13. (a) Describe about the various addressing modes.

Or

(b) Explain the General Register Organization in detail.

14. (a) Explain the need for communication link.

Or

(b) Explain about the priority interrupt.

15. (a) Describe the Memory Hierchy in detail.

Or

(b) Explain any two Auxillary storage devices.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the categories of computer Instructions.

17. Describe the Assembler passes in detail.

18. Explain about Instruction formats.

19. Explain with flow chart both multiplication algorithm.

20. Explain the principle of cache memory in detail.

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AFC–7458

Sub. Code
1BCAE1A

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Applications

Elective – WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the names of languages used in web design technology.

2. Write the syntax of the <FORM> tag.

3. What are the uses of scripting languages?

4. Write the syntax for the while structure.

5. How will you declare an Array in Java script?

6. What is the use of Boolean object?

7. What are the differences between static HTML and Dynamic HTML?

8. What is meant by object referencing?

9. Write the syntax for BGSOUND element

10. What is meant by scaling?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) How will you create the Lists in HTML?

Or

(b) Explain any five HTML tags with syntax and example.

12. (a) Explain about the Increment and decrement operators used in Java script.

Or

(b) Explain about the break and continue statement.

13. (a) Write about the Multiple subscripted Arrays.

Or

(b) Explain about the Programmer defined function.

14. (a) Explain about wave filter.

Or

(b) Explain about the dynamic styles.

15. (a) Write the syntax of EMBED element with example.

Or

(b) Explain about shape Primitives.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Design a form using HTML for college Admission.

17. Explain about switch structure and Do/while structure with example.

18. Explain about date object in Java script with an example.

19. Explain about Flip Filter and chroma filter.

20. Describe about direct animation.

AFC–7459

Sub. Code
1BCAE1B

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Applications

Elective — MOBILE COMMUNICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the use of Low Frequency? What is its range?

2. What is meant Dynamic Channel allocation?

3. What do you mean by Exposed terminals?

4. What is the role of Radio Subsystem (RSS)?

5. What are the advantages of Wireless LAN?

6. What do you mean by Handover?

7. What is Encapsulation in Mobile IP?

8. What are the advantages of selective retransmission?

9. What is meant by Reintegration?

10. Define POSTing.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain any Two Multiplexing schemes with a diagram.

Or

(b) Explain the advantages of Cellular systems.

12. (a) Write short notes on SDMA.

Or

(b) Explain the Four layers of DECT.

13. (a) Explain the operation of AD-HOC Networks.

Or

(b) Explain the Radio Access layer.

14. (a) Explain the functions Mobile TCP.

Or

(b) Explain Snooping.

15. (a) Explain the WWW system architecture.

Or

(b) Explain the applications of WAP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the various types of Modulation with necessary diagrams.

17. Describe the Digital Video Broadcasting.

18. Explain the MAC layer and its functions of Wireless LAN.

19. Explain the routing schemes for Mobile Ad-Hoc networks.

20. Explain the various file systems.

AFC–7460

Sub. Code
1BCAE2A

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Computer Application

***Elective* — COMPUTER GRAPHICS**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is Frame Buffer?

2. Define Aspect Ratio.

3. What do you mean by Geometric Transformation?

4. What is Reflection transformation?

5. Define Viewport.

6. What is meant by Clipping?

7. What is the difference between 2D and 3D transformation?

8. What are the 3D basic transformations?

9. Define Interface.

10. What is Command Language?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the working of CRT.

Or

(b) Write the Ellipse generation algorithm.

12. (a) Explain Composite Transformation with an example.

Or

(b) Explain the Transformation principles.

13. (a) Explain the concave polygon with a diagram.

Or

(b) Explain Line clipping with an example.

14. (a) Discuss 3D Translation transformation with an example.

Or

(b) Explain 3D Scaling transformation with an example.

15. (a) Explain the styles of command language.

Or

(b) Write short notes on Feedback.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write and explain the Bresenham's line drawing algorithm.

17. Explain the 2D Scaling, Rotation and Translation transformation with example.

18. Write and explain Sutherland Hodgman polygon clipping algorithm.

19. Explain the 3D Rotation transformation with example.

20. Explain the components of User Interface.

AFC–7461

Sub. Code
1BCAE2B

B.C.A DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Applications

***Elective* — LINUX PROGRAMMING**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is process in Linux?

2. What is inode?

3. What is Environment variables?

4. What is absolute path? Give an example.

5. What is redirection? Give an example.

6. What is background process?

7. What is positional parameters?

8. What is meta character?

9. What is file test?

10. What is ftp?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the features of Linux.

Or

(b) Discuss any five system calls in Linux.

12. (a) Explain any five simple shell commands.

Or

(b) Explain any five commands for navigating file system.

13. (a) Briefly explain about Grep utility.

Or

(b) Explain about uniq and tr filters.

14. (a) How do you carry out arithmetic in shell scripts?

Or

(b) Explain the different modes of operations in vi editor.

15. (a) Compare care control with multiple if structure in shell scripts.

Or

(b) Explain about Debugging and executing multiple scripts.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss process management in Linux.

17. What are basic file attributes? How to change the attributes?

18. Explain about process priorities and process scheduling.

19. How do write shell procedures? Give an example.

20. Explain about Linux system administration.

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AFC–7462

Sub. Code
1BCA6C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Applications

DATA MINING AND DATA WARE HOUSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is data warehousing?

2. What is meant by transforming data?

3. What are the tools needed to manage data warehouse?

4. What is process?

5. What is KDD?

6. What are the metrics used in Data mining?

7. What is OLTP system?

8. Define Genetic Algorithm.

9. What is Item set?

10. What is association rules?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the functions of Load Manager?

Or

(b) Explain about warehouse delivery method.

12. (a) What are the functions of warehouse manager?

Or

(b) How to estimate the Load?

13. (a) Explain the Basics of Data mining.

Or

(b) What are the social implications of data mining? Explain.

14. (a) Briefly discuss about Fuzzy Logic.

Or

(b) What is Dimensional modeling? Explain.

15. (a) How do you measure the quality of Rules?

Or

(b) Explain about incremental rules.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss process flow in a data ware house.

17. Explain Tuning the data load.

18. Explain about Data Mining issues.

19. Briefly discuss about Decision Trees and Neural Networks.

20. Explain parallel Algorithm for Association rules.

AFC–7463

Sub. Code
1BCA6C2

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Application

COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the uses of Computer Networks?

2. What is meant by Network standardization?

3. What are the design issues of Data Link Layer?

4. What do you mean by Protocol?

5. What is the use of Firewalls?

6. What is meant by Multicasting?

7. What is the basic function of Transport layer?

8. What do you mean by Multiplexing?

9. What is Cryptography?

10. What is Multimedia?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain various types of Network Topology.

Or

(b) Explain the working of Telephone system with a diagram.

12. (a) Explain the working of Sliding window protocol.

Or

(b) Explain the Petri net models.

13. (a) Explain any one congestion control algorithm.

Or

(b) Explain the Design issues of Network layer.

14. (a) Explain how connections are released by Transport layer protocols.

Or

(b) Explain TCP congestion control.

15. (a) Write short notes on SNMP.

Or

(b) Explain the JPEG and MPEG standards.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the functions of various layers in OSI reference model with a diagram.

17. Explain Carrier Sense Multiple Access protocols.

18. Explain various types of Routing algorithms.

19. Describe the Performance issues of Transport layer.

20. Describe the public key algorithms.

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AFC–7464

Sub. Code
1BCA6C3

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Applications

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define the term "Programmer".

2. What are the project size catagories?

3. Which principle is used in Expert Judgement Cost estimation?

4. What are formal specification techniques?

5. What is meant by milestone?

6. Define – Real time system.

7. Define – unit testing.

8. Write the art of debugging.

9. What is meant by walk through?

10. Define "Software Review".

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Compare Trivial Projects and Small Projects.

Or

(b) Describe Managerial Issues in detail.

12. (a) Describe the software cost factors.

Or

(b) Explain Delphi Cost-estimation.

13. (a) Describe the distributed system design.

Or

(b) Describe the design notations.

14. (a) Describe the configuration management in detail.

Or

(b) Explain the validation testing.

15. (a) Explain the ISO 9000 quality standards.

Or

(b) Explain the software quality assurance.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the planning an organization structure.
17. Explain the staffing level estimation.
18. Explain inspections in detail.
19. Explain the managerial aspects of software engineering.
20. Describe formal technical reviews.

AFC–7604

Sub. Code
1BCASA1/ 1BCA1A1

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Applications – Allied

DISCRETE MATHEMATICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define Disjunction.

2. Construct the truth table for $p \leftrightarrow q$.

3. Obtain the PDNF for $\neg P \vee Q$.

4. What do you mean by quantifier?

5. Define degree of a graph.

6. What is a complete graph?

7. Define spanning tree.

8. What is cut vertices?

9. Define partition and covering of a set.

10. What is lattice?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain disjunction with example and truth table.

Or

(b) Verify whether $(p \vee q) \vee \neg(p \wedge q)$ is a tautology.

12. (a) Obtain a disjunctive normal form of $\neg(P \vee Q) \leftrightarrow (P \wedge Q)$.

Or

(b) Show that

$(\forall x)\, P(x)\, V\, Q(x)) \Rightarrow (\forall x)\, P(x)\, V\, (\exists x)\, Q(x)$.

13. (a) Let G be a graph. Then $\sum_{u \in V} d(V) = 2e$, where $e = |E(G)|$.

Or

(b) Explain isomorphic graphs with examples.

14. (a) Explain matrix tree theorem.

Or

(b) Explain Hamiltonian graph.

15. (a) Write about equivalence relations.

Or

(b) Write about special lattices.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove that $(P \to (Q \to R)) \Rightarrow ((P \to Q) \to (P \to R))$.

17. Write a detailed note on theory of inference for predicate calculus.

18. Explain :

    (a) Subgraph

    (b) Bipartite graph.

19. Explain Kurskal's algorithm.

20. Explain about Boolean functions.

AFC–7605

Sub. Code
1BCASA2

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Allied – Computer Applications

RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define a model.

2. What are the phases of scientific method in OR?

3. Define slack variable.

4. In the two-phase method explain when Phase-1 terminates?

5. Define the dual of a LPP;

   Min $z = C^T x$ subject to

   $Ax = b,\ x \geq 0$

6. Define Integer programming problem.

7. Define Assignment problem.

8. What is travelling salesman problem?

9. Show that assignment problem is the special case of the transportation problem.

10. What is an unbalanced transportation problem?

Part B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

11. (a) Give various definitions of OR.

Or

(b) Discuss the various phases in solving an OR problem.

12. (a) Solve the following LP problem graphically.

Max $z = x_1 + x_2$, subject to

$$x_1 + 2x_2 \leq 2000$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

and $x_1, x_2 \geq 0$

Or

(b) Write the computational procedure for simplex method.

13. (a) Use dual simplex method to solve the following LPP.

Max $z = -3x_1 - 2x_2$

Subject to $x_1 + x_2 \geq 1$

$$x_1 + x_2 \leq 7$$

$$x_1 + 2x_2 \leq 10$$

$$x_2 \leq 3, \; x_1, x_2 \geq 0$$

Or

(b) Explain Gomary's cutting-plane Algorithm.

14. (a) Explain the computational procedure for Hungarian Assignment method.

Or

(b) Solve the following assignment problem.

Subordinates

Tasks	I	II	III	IV
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

15. (a) Determine the initial basic feasible solution of TP using vogel's method.

		Destination				Supply
		1	2	3	4	
Source	1	1	2	1	4	30
	2	3	3	2	1	50
	3	4	2	5	9	20
Demand		20	40	30	10	100

Or

(b) Explain the procedure for obtaining optimal solution in transportation problem.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss the scope of operations research.

17. Use two-phase simplex method to solve the problem.

Minimize $z = x_1 - 2x_2 - 3x_3$

Subject to the constraints

$-2x_1 + x_2 + 3x_3 = 2$,

$2x_1 + 3x_2 + 4x_3 = 1$ and $x_1, x_2 \geq 0$

18. Find the optimum integer solution to the following all integer programming problem.

Max $z = x_1 - 2x_2$

Subject to $4x_1 + 2x_2 \leq 15$

$x_1 \geq 0,\ x_2 \geq 0$ and integers.

19. Solve the travelling–salesman problem given by the following data.

$C_{12} = 20$, $C_{13} = 4$, $C_{14} = 10$, $C_{23} = 5$

$C_{34} = 6$, $C_{25} = 10$, $C_{35} = 6$, $C_{45} = 20$

Where $C_{ij} = C_{ji}$ and there is no route between cities i and j if the value for C_{ij} is not shown.

20. Find the optimum solution to the following transportation problem.

		Store			a_i
		A	B	C	
Factory	I	10	9	8	8
	II	10	7	10	7
	III	11	9	7	9
	IV	12	14	10	4
$b_j \rightarrow$		10	10	8	

AFC–7606

Sub. Code
1BCASA3

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Applications — Allied

ACCOUNTING PRINCIPLE AND COMPUTER APPLICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. What do you mean by accounting principles?

2. Define — Accounting

3. What is a compound entry?

4. What do you mean by double entry system?

5. What is petty cash?

6. What is a cash book?

7. What are bad debts?

8. What is closing stock?

9. What is meant by petty cash book?

10. What is bank Reconciliation Statement?

Part B (5 × 3 = 15)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the objectives of book keeping?

Or

(b) Explain Going Concern Concept.

12. (a) Why is journal called the book of original entry?

Or

(b) What are the advantages of Trial Balance?

13. (a) Enter the following transactions in the purchases return book of Mr. Arun

2012

March

1 Returned to Nellai Stores Goods worth Rs. 750

8 Returned 10 boxes of soaps to Colombo stores Rs. 1,500

25 Returned to Modem Culture Palace 10 Thanjavur Plates @ 200 each, subject to 20% Trade discount.

Or

(b) What are the merits of subsidiary books?

14. (a) From the information is given below prepare trading account.

	Rs.
Opening stock	1,00,000
Purchases	1,50,000
Purchase returns	25,000
Direct expenses	10,000
Carriage inwards	5,000
Sales	4,00,000
Closing stock	50,000

Or

(b) What are the uses of balance sheet?

15. (a) How will you prepare a bank reconciliation statement?

Or

(b) On 30.9.2012 the cash book Mr. Gopal showed a favourable bank balance of Rs. 2,500. This balance did not tally with the balance shown by the pass book.

From the following particulars ascertain the bank balance as per pass book.

(i) Cheques sent to the bank for collection but not credited up to 30.9.20 12 Rs. 500

(ii) Cheques issued to customer but not presented for payment up to 30.9.20 12 Rs. 675

(iii) Bank charges not entered in the cash book Rs. 10

(iv) A cheque issued by Mr. Gopalasamy for Rs. 120 has been wrongly debited to Gopal's account by the bank.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Journalise the following transactions :

2012 Jan.	2	Started business with Rs. 10,000; Paid into bank Rs. 5,000
	3	Bought furniture Rs. 500
	4	Bought goods for Rs. 3,000
	5	Sold goods for Rs. 600
	6	Paid telephone Rent Rs. 240
	7	Paid Rs. 100 for Advertisement
	8	Sold goods to Mr. S. Lal & Co., for Rs. 800 for cash
	9	Withdrew from bank Rs. 300 for private use
	10	Electric charges amounted Rs. 120
	11	Received commission Rs. 240.

17. From the following transactions, prepare ledger accounts

Feb.	1	Sold goods for cash Rs. 1,300
	2	Purchased goods Rs. 400
	3	Purchased goods from Kumar Rs. 3,000
	4	Sold goods to Mr. Prabu Rs. 2,000
	5	Received cash from Mr. Prabu Rs. 1,200
	6	Paid to Mr. Kumar Rs. 1,000
	7	Paid Salary Rs. 700.

18. Prepare a double column cash book from the following transactions of Mr. Raja

April 1	Cash in hand Rs. 4,000
6	Cash purchases Rs. 2,000
10	Wages paid Rs. 40
11	Cash sales Rs. 6,000
12	Cash received from Mr. Suresh Rs. 1,980 allowed him discount Rs. 20
19	Cash paid to Ms. Meena Rs. 2,470 and discount received Rs. 30
27	Cash paid to Ms. Radha Rs. 400
28	Purchased goods for cash Rs. 2,070.

19. The following is the balance of a person on 31st December 2010

Debit	Rs.	Credit	Rs.
Cash in hand	540	Sales	98,780
Cash at bank	2,630	Returns out	500
Purchases	40,675	Capital	71,000
Returns	680	Creditors	6,300
Wages	10,480		
Fuel	4,730		
Carriage outwards	3,200		
Carriage inwards	2,040		
Opening stock	5,760		
Premises	30,000		
Land	10,000		
Machinery	20,000		
Patents	7,500		

Debit	Rs.	Credit	Rs.
Salaries	15,000		
Sundry expenses	3,000		
Insurance	600		
Drawings	5,245		
Debtors	14,500		
	1,76,580		1,76,580

Closing stock Rs. 16,800

Prepare Trading, Profit and Loss account and Balance sheet.

20. From the following particulars prepare a bank reconciliation statement as on 31st December 2011

(a) Bank balance as per cash book as on 3 1.12.2011, Rs. 5,400

(b) Cheque issued but not presented for payment Rs. 515

(c) Cheques deposited in to bank but not credited in the pass book Rs. 240

(d) Bank charges is debited in pass book Rs. 30

(e) Interest on current account credited by the bank but not recorded in the cashbook Rs. 25.

AFC–7607

Sub. Code
1BCASA4

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Applications — Allied

COST AND MANAGEMENT ACCOUNTING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. Define - Cost accounting.

2. What is meant by semi-variable expenses?

3. What is meant by average stock level?

4. What is danger level?

5. What do you mean by incentive wage payment?

6. Define – Over Time.

7. What is P/V ratio?

8. What is meant by key factor?

9. What do you mean by master budget?

10. What do you mean by budget manual?

Part B (5 × 3 = 15)

Answer the following questions, choosing either (a) or (b).

11. (a) What are the elements of cost accounting?

Or

(b) Calculate prime cost, factory cost, cost of production, cost of sales and profit from the following details :

Direct Materials Rs. 10,000

Direct Labour Rs. 4,000

Direct Expenses Rs. 500

Factory Expenses Rs. 1,500

Administrative Expenses Rs. 1,000

Selling Expenses Rs. 300

Sales Rs. 20,000.

12. (a) Write short notes on waste, scrap and spoilage.

Or

(b) Find out EOQ from the following :

Annual usage 6,000 units, cost of material per unit Rs. 20, cost of placing and receiving one order Rs. 60, annual carrying cost of one unit 10% of inventory.

13. (a) Raghavendra Metal Company gives the following information :

No. of employees on 1.4.2011	200
No. of employees on 3 1.3.2012	240
No. of employees resigned	20
No. of employees discharged	5
No. of employees replaced	18

Calculate labour turnover rate by applying three methods.

Or

(b) What are the essentials of a good wage system?

14. (a) The following information is given to you.

Sales Rs. 4,00,000

Fixed Cost Rs. 1,80,000

Variable Cost Rs. 2,70,000

Calculate break even sales.

Or

(b) What are the limitations of marginal costing techniques?

15. (a) What do you understand by zero base budgeting?

Or

(b) A company expects to sell 50,000 units of a certain product over the next year. Two kinds of materials A and B are required to make the product. Each unit of the product requires 2 units of A and 3 units of B. The opening and closing stocks for the year estimated to be as follows :

Particulars	Opening stock Units	Closing stock Units
Finished product	10,000	10,000
Materials :		
A	12,000	13,000
B	15,000	16,000

Draw up a purchase budget.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What are the differences between cost accounting and financial accounting?

17. The following are the extract of record of receives and issues during 2012 :

Jan 1 Opening balance 500 tones @ Rs. 2 per tones

Jan 8 Issued 250 tones

Jan 13 Received from supplier 200 tones @ Rs. 1.90

Jan 16 Issued 180 tones

Jan 20 Received from supplier 240 tones @ Rs. 1.80

Jan 24 Issued 300 tones

Jan 25 Received from supplier 320 tones @ Rs. 1.90

Jan 28 Issued 200 tones

Jan 29 Return to stores 30 tones @ Rs. 1.90

Issues are to be priced on the principle of FIFO method. Prepare stores ledger.

18. From the following particulars given below prepare Labour cost per day of 8 hours.

(a) Basic Salary Rs. 2 per day

(b) D.A Rs. 25 per every point over 100 cost of living index for working class. Current cost of living index 700 points

(c) Leave salary 10% of item (a) & (b)

(d) Employer's contribution of P.F 8% of item (a), (b) and (c)

(e) Employer's contribution to ESI 2.5% of item (a), (b) and (c)

(f) Expenditure of amenities to worker Rs. 20 per head per month

(g) Number of working days in a month 25 days of 8 hours each.

19. The operating results of a company for the last 2 years are as follows :

Particulars	Sales	Profit
2010	2,70,000	6,000
2011	3,00,000	15,000

You are required to calculate,

(a) P/V ratio

(b) Fixed Cost

(c) Margin of Safety at a profit of Rs. 24,000.

20. Happy Ltd., have prepared the flexible budget for the production of 10,000 units of a commodity manufacture by them.

Raw Materials	2.52
Direct labour	0.75
Direct expenses	0.10
Works overhead (60% fixed)	2.50
Administration overhead (80% fixed)	0.40
Selling overhead (50% fixed)	0.20

Prepare a revised flexible budget for the production of 6,000 and 7,000 units, showing total cost and cost per unit.

AFC–7608

Sub. Code
1BCAA1/ 1BCA1A2

U.G DEGREE EXAMINATION, NOVEMBER 2015

Computer Application – Allied

COMPUTER FUNDAMENTALS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a pixel?

2. What are the characteristics of a Computer?

3. What is the purpose of Secondary memory?

4. What is a Digitizer?

5. Define Algorithm.

6. What is the difference between Compiler and an Interpreter?

7. What is an Operating System?

8. What is the difference between executable and a non-executable file?

9. What is an Icon?

10. What is the use of Notepad?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the functions of various components of a Computer with a block diagram.

Or

(b) Explain the features of a Super Computer.

12. (a) Write short notes on LCD.

Or

(b) Explain the working of a Scanner.

13. (a) Explain the various types of programming languages.

Or

(b) Explain the representation of an algorithm with an example.

14. (a) Explain the Booting process.

Or

(b) Explain any Five DOS commands with example.

15. (a) Explain the various Windows applications.

Or

(b) Write short notes on Task Bar.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain (a) CPU (b) Keyboard.

17. Explain various types of ROMs.

18. Explain the symbols used in Flow Chart. Explain the rules and advantages of Flow Chart.

19. Explain various types of Operating system.

20. Explain (a) Control Panel (b) Windows Explorer.

AFC–7452

Sub. Code
1BCA1C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Application

PROGRAMMING IN C

(CBCS – 2011 onwards)

Time : 3 Hours Maximum :75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define a constant. How will you declare a constant in C.

2. Write any four types of expressions in C with example.

3. Write any four string handling functions in C.

4. How will you initialize a Two dimensional array?

5. What is the need for user defined functions in C?

6. Write any Two differences between structure and union.

7. What are the advantages of a pointer?

8. Write a pointer expression to multiply two numbers.

9. How will you open a file?

10. Define Macro.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain various input statements in C with example.

Or

(b) Explain the operators in C with example.

12. (a) Write a program to reverse the given string.

Or

(b) Write a program to count the number of uppercase letters in a word.

13. (a) Explain recursion with a program.

Or

(b) Explain how arrays can be used within a structure? Give example.

14. (a) Explain with an example how will you declare and initialize a pointer?

Or

(b) Explain with an example how will you assign pointers to a structure?

15. (a) Explain various file opening modes with example.

Or

(b) Explain any five compiler control directives.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Write a program to print the first ten Fibonacci series using (a) while loop (b) do-while loop.

17. Write a program to find the sum of the elements of a given matrix.

18. Write a program to find the total mark obtained by a student using array of structures.

19. Write a program to arrange the given set of numbers using pointers.

20. Write a program to store the details of the employee in a file. Write another program to read the details of the employee whose basic pay is less than Rs 5000/-.